

Entity-Relationship Diagram - Gym Management System

1. User & Authentication Entities

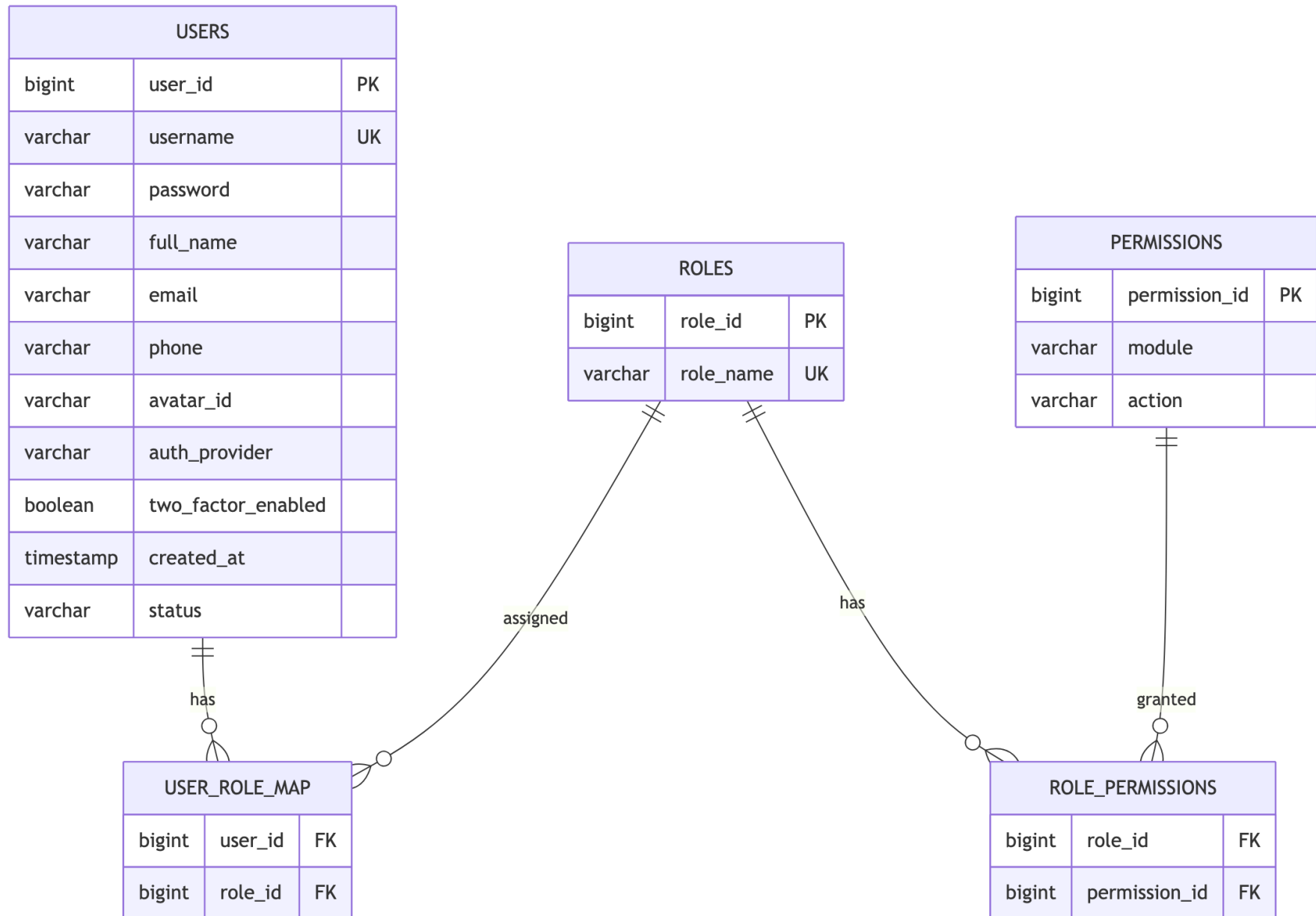
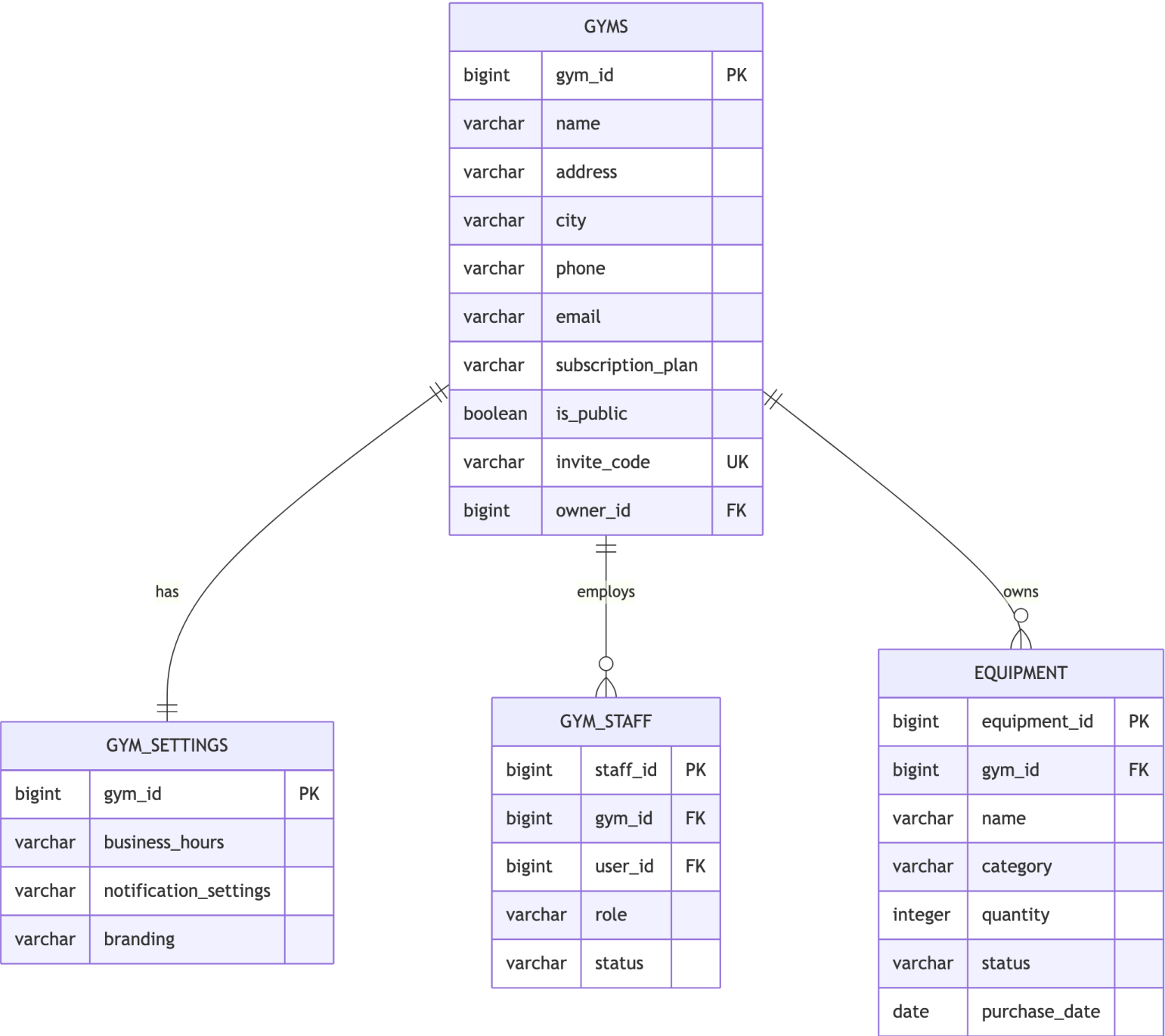


Figure 1: Diagram

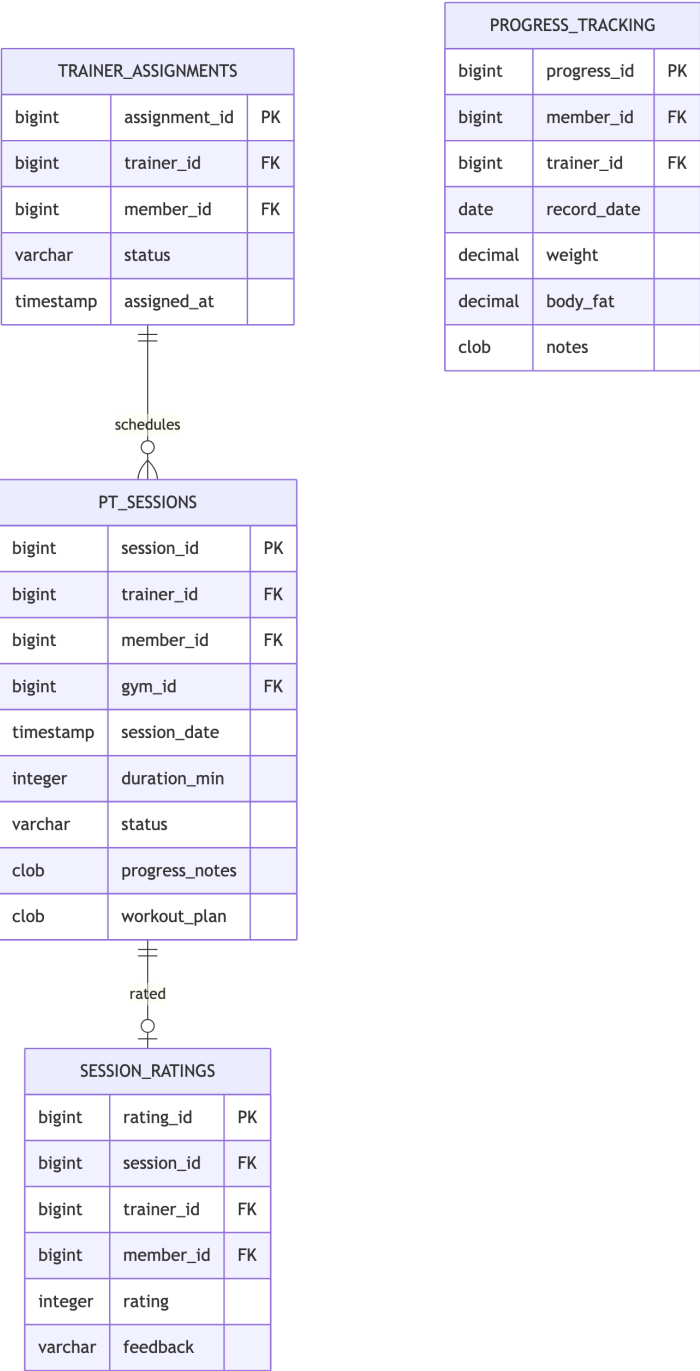
2. Gym Management Entities



3. Membership Entities



4. Training & Session Entities



5. Communication Entities

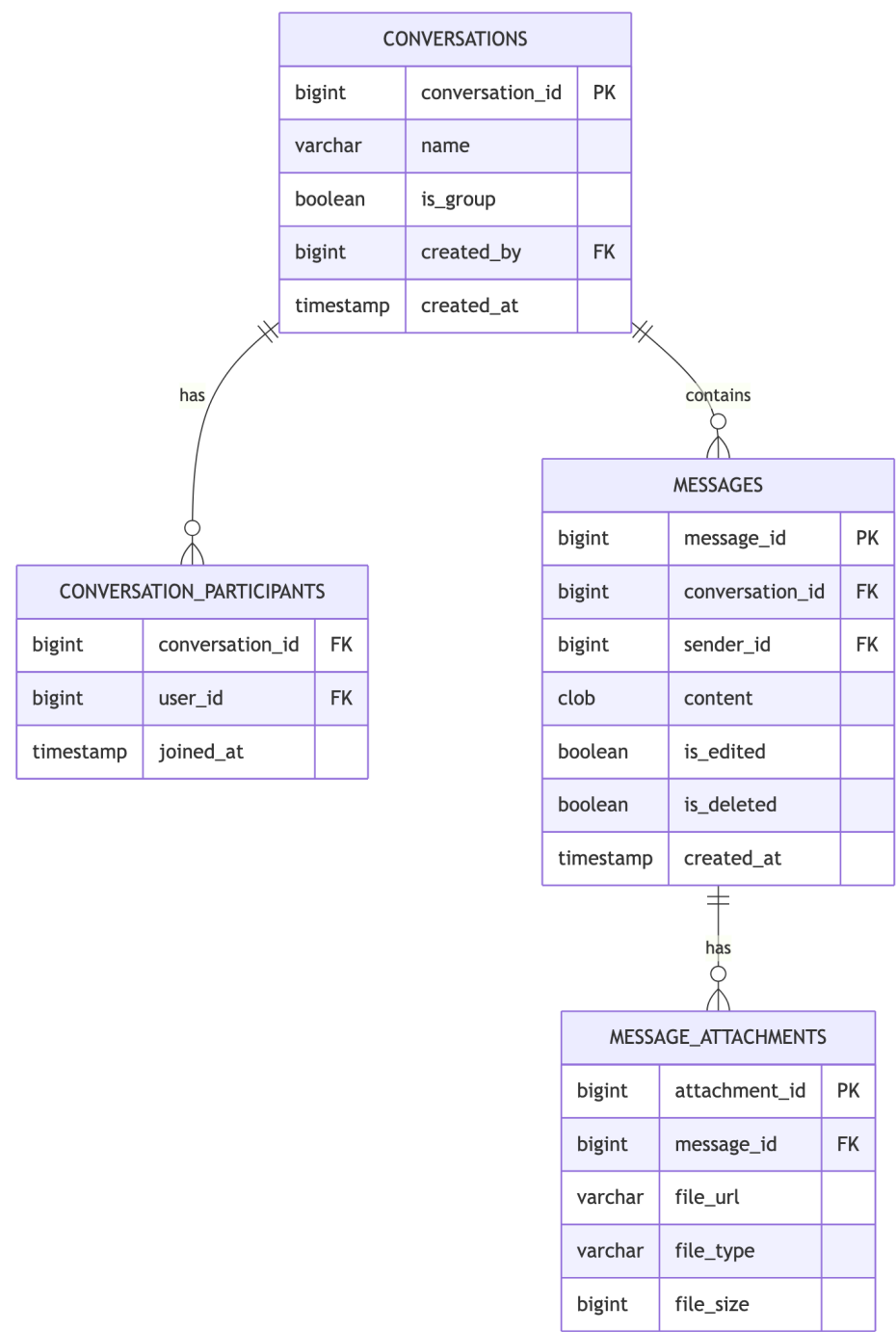


Figure 5: Diagram

6. Notification Entities

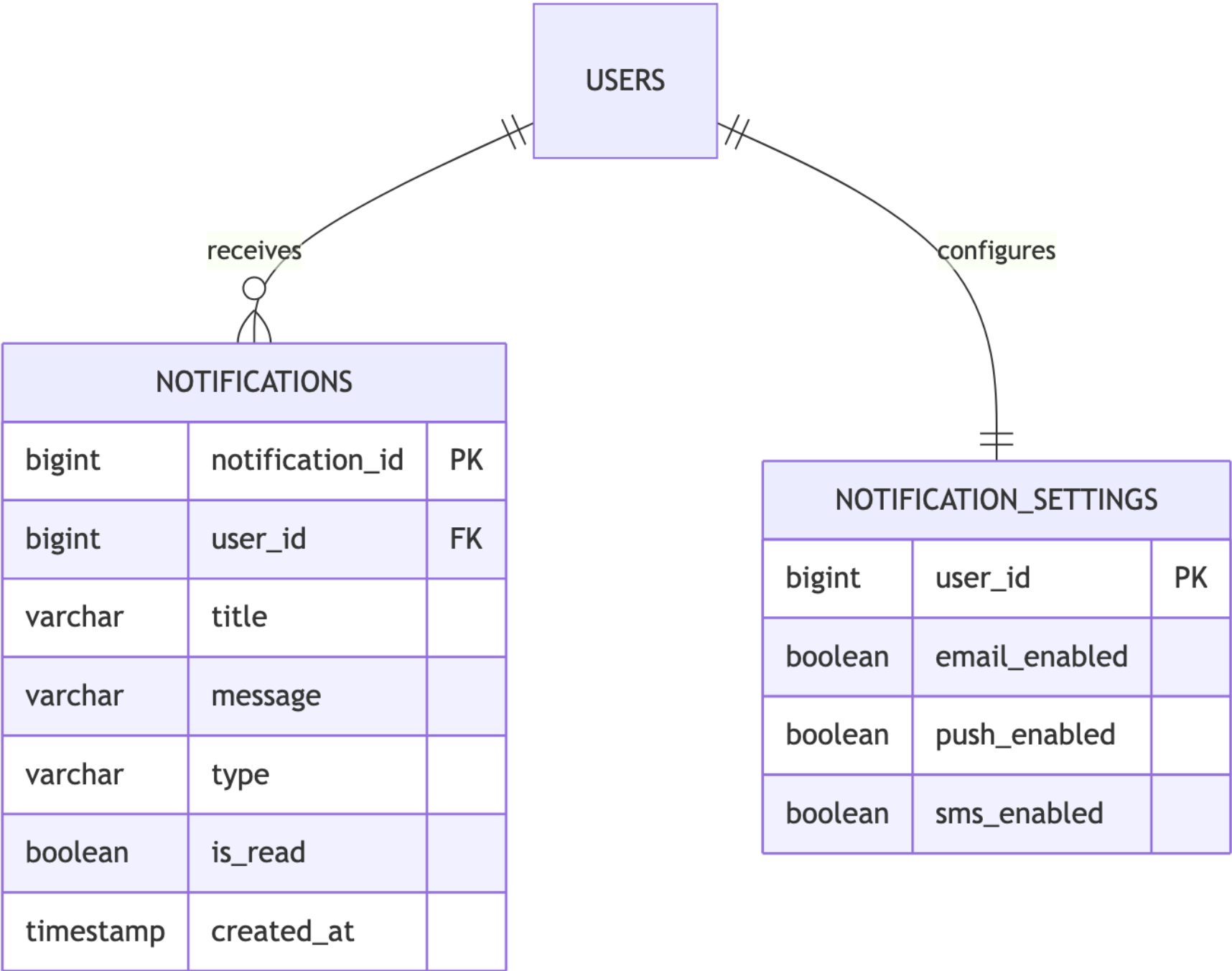
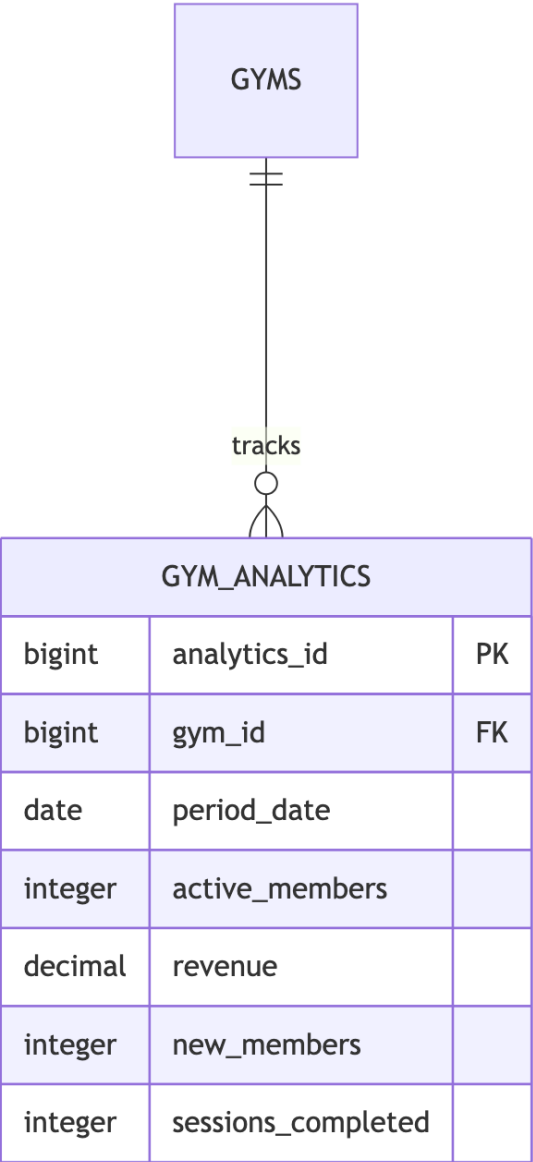


Figure 6: Diagram

7. Analytics Entities

TRAINER_STATS		
bigint	stats_id	PK
bigint	trainer_id	FK
date	period_date	
integer	sessions_count	
decimal	avg_rating	
integer	active_clients	



Entity Summary Table

Category	Entities	Key Relationships
Auth	Users, Roles, Permissions	Many-to-many via mapping tables
Gym	Gyms, Settings, Staff, Equipment	Owner-owned, one-to-many
Members	Packages, Memberships, Transactions	Package □ Membership □ Transaction
Training	Assignments, Sessions, Ratings, Progress	Trainer-Member assignments
Chat	Conversations, Messages, Attachments	Conversation hierarchy
Notify	Notifications, Settings	User notifications
Analytics	Gym Analytics, Trainer Stats	Period-based aggregations